

Study Guide — RAID Storage Technology

Overview

This chapter focuses on:

- RAID fundamentals
- Need for RAID
- Data loss and disk failures
- RAID concepts
- Striping
- Mirroring
- Parity
- RAID levels
- RAID 0, 1, 5, 6, and 10
- Cloud RAID concepts
- RAID performance tuning
- Caching techniques
- Advantages of RAID

This guide teaches every concept step-by-step with explanations and examples.

1. What is RAID?

RAID stands for:

Redundant Array of Independent Disks

RAID combines multiple physical drives into:

- one logical drive.

The operating system sees:

- a single storage unit.
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2. Purpose of RAID

RAID is used to improve:

- storage capacity
- performance
- reliability
- fault tolerance

3. Why RAID is Important

Storage devices fail over time.

Studies show:

- around 1.5%–5% of drives fail annually.

In large data centers:

- failures happen frequently.

Without RAID:

- one disk failure may cause total data loss.
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4. Real-World Importance

Cloud providers and data centers use RAID because:

- millions of users depend on continuous access
 - downtime is expensive
 - data must remain available even during failures
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5. RAID Technology Basics

RAID distributes data across multiple disks.

Different RAID configurations provide:

- higher speed
 - redundancy
 - fault tolerance
 - or combinations of these
-

6. RAID Configurations

RAID configurations are represented using numbers:

- RAID 0
- RAID 1
- RAID 5

- RAID 6
- RAID 10

Each level has different:

- speed
 - storage efficiency
 - fault tolerance
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7. Key RAID Concepts

Three major concepts:

- Striping
 - Mirroring
 - Parity
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8. Striping

Striping means:

splitting data into blocks and distributing them across disks.

Used in:

- RAID 0
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Advantages of Striping

- Faster read speed
 - Faster write speed
 - Parallel disk access
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Problem with Striping

No redundancy.

If one disk fails:

- all data lost.
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9. Mirroring

Mirroring means:

creating exact duplicate copies of data.

Used in:

- RAID 1
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Advantages of Mirroring

- Excellent fault tolerance
 - High reliability
 - Fast recovery
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Disadvantage of Mirroring

Storage efficiency reduced.

Example:

- 2TB total disks
- only 1TB usable

because duplicate copy stored.

10. Parity

Parity uses mathematical calculations to recover lost data.

Used in:

- RAID 5
 - RAID 6
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11. XOR Operation in Parity

Parity commonly uses:

XOR (Exclusive OR)

Example:

Disk 1: 1 0 1 1

Disk 2: 0 1 1 0

Parity: 1 1 0 1

12. How Parity Recovers Data

If one disk fails:

Remaining disk XOR parity = lost data.

This allows:

- data reconstruction.
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13. RAID 0

RAID 0 uses:

- striping only

No mirroring. No parity.

Advantages of RAID 0

- highest performance
 - fast reads/writes
 - full storage utilization
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Disadvantages of RAID 0

No fault tolerance.

If one disk fails:

- entire array fails.
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Best Use Cases

- gaming
 - temporary processing
 - non-critical high-speed workloads
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14. RAID 1

RAID 1 uses:

- mirroring

Data duplicated across disks.

Advantages of RAID 1

- excellent reliability
 - fast recovery
 - high availability
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Disadvantages of RAID 1

- expensive
 - only 50% storage usable
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Best Use Cases

- operating systems
 - critical servers
 - important databases
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15. RAID 5

RAID 5 uses:

- striping
- distributed parity

Requires:

- minimum 3 disks
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Advantages of RAID 5

- good performance
 - fault tolerance
 - efficient storage usage
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Fault Tolerance in RAID 5

Can survive:

- 1 disk failure
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Disadvantages of RAID 5

- parity calculations create overhead
 - rebuild time can be slow
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Best Use Cases

- file servers
 - enterprise storage
 - application servers
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16. RAID 6

RAID 6 uses:

- striping
- dual parity

Requires:

- minimum 4 disks
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Advantages of RAID 6

Can survive:

- 2 simultaneous disk failures

Very reliable.

Disadvantages of RAID 6

- slower writes
 - more parity overhead
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Best Use Cases

- large storage arrays
 - critical enterprise systems
 - backup systems
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17. RAID 10

RAID 10 combines:

- RAID 1 (mirroring)
- RAID 0 (striping)

Also written as:

RAID 1+0

18. Features of RAID 10

Provides:

- high performance
 - high redundancy
 - fast recovery
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Advantages of RAID 10

- excellent speed
 - excellent reliability
 - very fast rebuilds
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Disadvantages of RAID 10

- expensive
 - requires many disks
 - only half storage usable
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Best Use Cases

- MySQL databases
 - PostgreSQL databases
 - high-traffic systems
 - enterprise applications
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19. Cloud RAID Concepts

Cloud providers use replication concepts similar to RAID.

20. Azure Storage Replication Types

LRS (Locally Redundant Storage)

Stores:

- 3 copies
- same data center

Comparable to:

- local RAID redundancy
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ZRS (Zone Redundant Storage)

Stores:

- 3 copies
- different availability zones

Provides:

- zone-level fault tolerance
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GRS (Geo-Redundant Storage)

Stores:

- 6 copies
- across different regions/continents

Provides:

- disaster recovery
 - regional fault tolerance
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21. RAID Performance Tuning

Performance depends on:

- chunk size
 - caching
 - workload type
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22. Chunk Size

Chunk size determines:

how much data written per disk operation.

Small Chunk Size

Better for:

- databases
- small random operations

But:

- more overhead
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Large Chunk Size

Better for:

- video streaming
- large sequential reads

Less overhead.

23. Read Ahead Caching

Storage systems predict future reads.

Example:

If user reads Block 5:

System predicts:

- Block 6
- Block 7
- Block 8

These blocks loaded into cache.

Benefits of Read Ahead Caching

- lower latency
 - faster reads
 - improved performance
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24. Write Through Cache

Data written:

- to cache
 - and immediately to disk
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Advantages

- safer
 - reliable
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Disadvantage

- slower performance
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25. Write Back Cache

Data first written to cache.

Disk updated later.

Advantages

- much faster writes
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Risk

Power failure may lose cached data.

26. Battery-Backed Cache

To solve cache risks:

systems use:

- battery-backed cache
- nvRAM

This preserves data during power loss.

27. Advantages of RAID

Improved Performance

Multiple disks work simultaneously.

Fault Tolerance

Data survives disk failures.

Higher Availability

Systems remain online.

Scalability

Storage capacity increases.

Reliability

Redundant storage protects data.

28. Limitations of RAID

RAID is NOT backup.

RAID cannot protect against:

- accidental deletion
 - malware
 - ransomware
 - corruption
 - fire/disasters (unless geo-redundant)
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29. RAID Levels Comparison

RAID Level	Technique	Fault Tolerance	Performance	Minimum Disks
RAID 0	Striping	None	Very High	2
RAID 1	Mirroring	1 Disk	High	2
RAID 5	Striping + Parity	1 Disk	Good	3
RAID 6	Striping + Dual Parity	2 Disks	Moderate	4
RAID 10	Mirroring + Striping	Multiple	Very High	4

30. Important Terms to Memorize

- RAID
- Striping
- Mirroring
- Parity
- XOR
- Fault tolerance
- Replication
- Chunk size
- Read ahead caching
- Write back cache
- Write through cache
- LRS

- ZRS
- GRS

31. Important Comparisons

RAID 0 vs RAID 1

RAID 0	RAID 1
Fastest	Reliable
No redundancy	Full redundancy
Data loss risk	Safe from single disk failure

RAID 5 vs RAID 6

RAID 5	RAID 6
Single parity	Dual parity
1 disk failure tolerated	2 disk failures tolerated
Faster writes	Slower writes

Write Through vs Write Back

Write Through	Write Back
Safer	Faster
Immediate disk write	Delayed disk write

32. Likely Exam Questions

1. What is RAID?
2. Why is RAID important?
3. Explain striping.
4. Explain mirroring.
5. Explain parity.
6. How does XOR work in RAID?
7. Compare RAID 0 and RAID 1.
8. Explain RAID 5.
9. Explain RAID 6.

10. Explain RAID 10.
 11. Compare RAID levels.
 12. Explain cloud RAID concepts.
 13. Explain read ahead caching.
 14. Compare write through and write back cache.
 15. Explain advantages and limitations of RAID.
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33. Quick Revision Notes

- RAID combines multiple disks.
 - RAID improves performance and reliability.
 - Striping increases speed.
 - Mirroring increases redundancy.
 - Parity enables recovery.
 - RAID 0 has no fault tolerance.
 - RAID 1 duplicates data.
 - RAID 5 uses single parity.
 - RAID 6 uses dual parity.
 - RAID 10 combines mirroring and striping.
 - Cloud replication is similar to RAID concepts.
 - Read ahead caching improves performance.
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Final Understanding

By the end of this chapter, you should understand:

- how RAID improves storage systems
- differences between RAID levels
- how striping, mirroring, and parity work
- how fault tolerance is achieved
- how cloud providers implement redundancy
- how caching improves RAID performance
- which RAID level is suitable for different workloads